

CSC577

SOFTWARE ENGINEERING: THEORIES AND PRINCIPLES

LECTURE: 031 – SOFTWARE PROCESSES



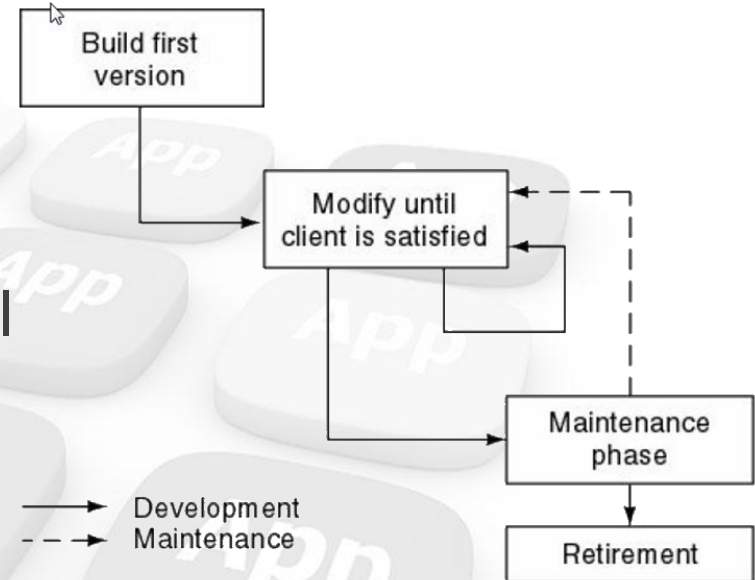
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TOPIC OUTCOME(S)

- To introduce software process models
- To describe three generic process models and when they may be used
- To describe outline process models for requirements engineering, software development, testing and evolution

BUILT AND FIX

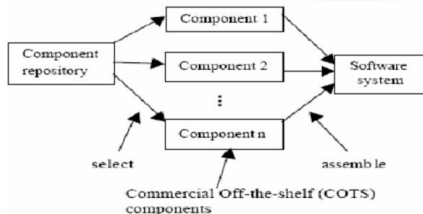
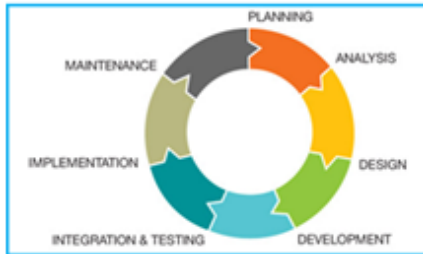
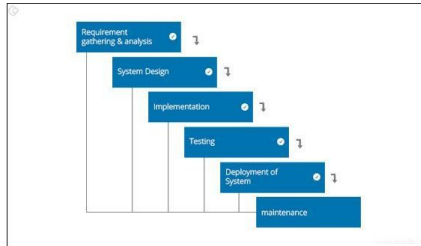
- Problems
 - No specifications
 - No design
- Totally unsatisfactory
- Need life-cycle model
 - "Game plan"
 - Phases
 - Deliverables & Milestones



SOFTWARE PROCESS

- A structured set of activities required to develop a software system
 - Analysis
 - Specification
 - Design
 - Design & Implementation
 - Development
 - Design & Implementation
 - Testing
 - Verification & Validation
 - Deployment
 - Evolution
 - Maintenance
 - Evolution

SDLC PHASES



Phase	Description
Specification	An explicit set of requirements of a material, design, product, or service.
Design	The creation of a plan, idea, model or convention for the construction of a system.
Implementation	The execution of a plan, idea, model or convention of a system.
Verification	The act of reviewing and inspecting in order to establish and document that a product, service or system meets regulatory or technical standards.
Validation	Checking that a software system meets specifications and fulfills its intended purpose
Evolution	Deployment & Maintenance

WHAT IS A SOFTWARE PROCESS MODEL? ^{6/28}

- A simplified representation of a software process, presented from a specific perspective
- Examples of process perspectives are
 - Workflow perspective - sequence of activities
 - Data-flow perspective - information flow
 - Role/action perspective - who does what

GENERIC SOFTWARE PROCESS MODELS

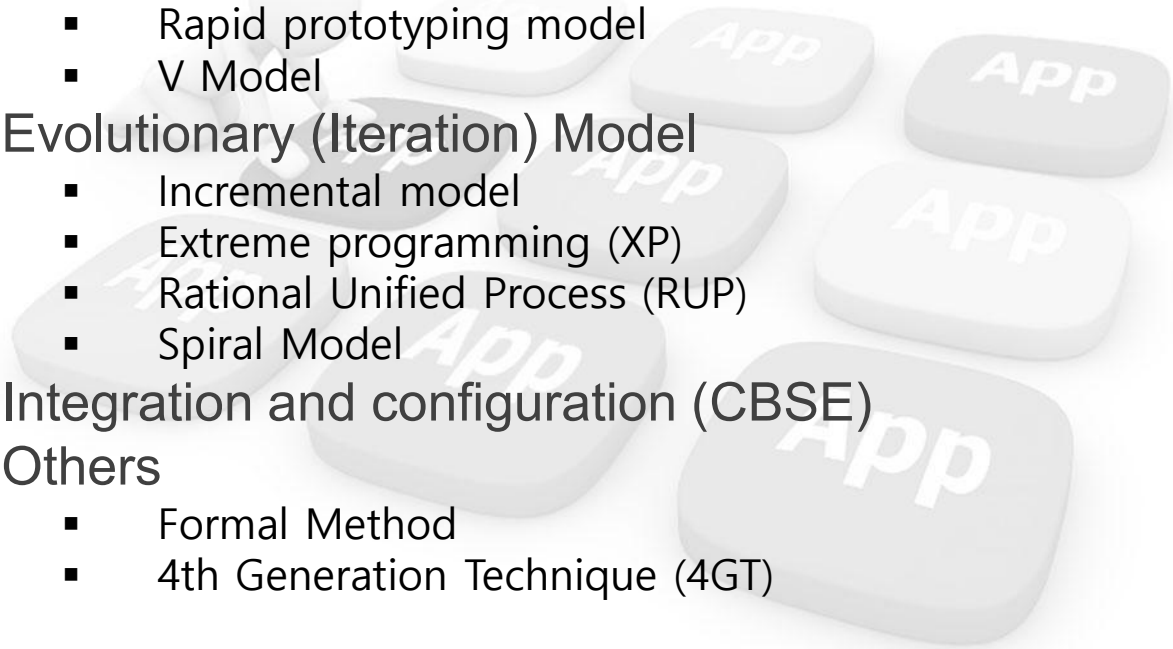
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- The waterfall (plan based) model
 - Separate and distinct phases of specification and development
- Evolutionary (iteration) model
 - Specification, development and validation are interleaved
- Integration and configuration (CBSE)
 - The system is assembled from existing components
- There are many variants of these models e.g. formal development where a waterfall-like process is used but the specification is a formal specification that is refined through several stages to an implementable design

SOFTWARE PROCESS MODELS — 8/28

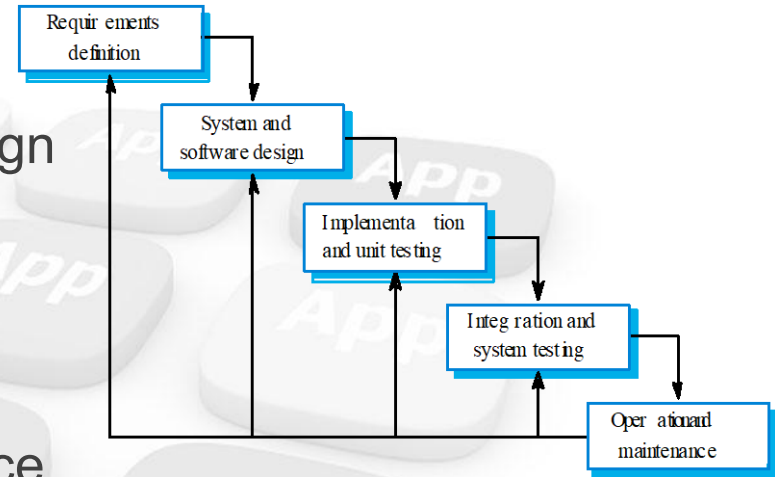
EXAMPLES

- Waterfall (Plan Based) Model
 - Waterfall Model
 - Rapid prototyping model
 - V Model
- Evolutionary (Iteration) Model
 - Incremental model
 - Extreme programming (XP)
 - Rational Unified Process (RUP)
 - Spiral Model
- Integration and configuration (CBSE)
- Others
 - Formal Method
 - 4th Generation Technique (4GT)



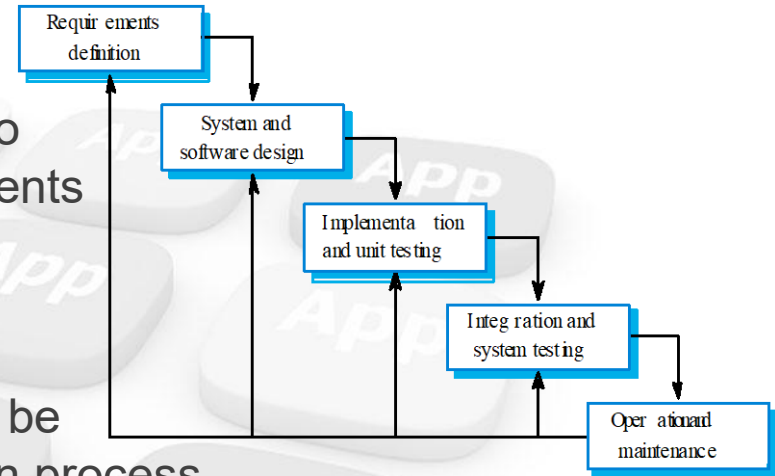
WATERFALL MODEL PHASES

- Requirements analysis and definition
- System and software design
- Implementation and unit testing
- Integration and system testing
- Operation and maintenance
- The main drawback of the waterfall model is the difficulty of accommodating change after the process is underway. One phase has to be complete before moving onto the next phase



WATERFALL MODEL PROBLEMS

- Inflexible partitioning of the project into distinct stages makes it difficult to respond to changing customer requirements
- Therefore, this model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design process
- Few business systems have stable requirements
- The waterfall model is mostly used for large systems engineering projects where a system is developed at several sites



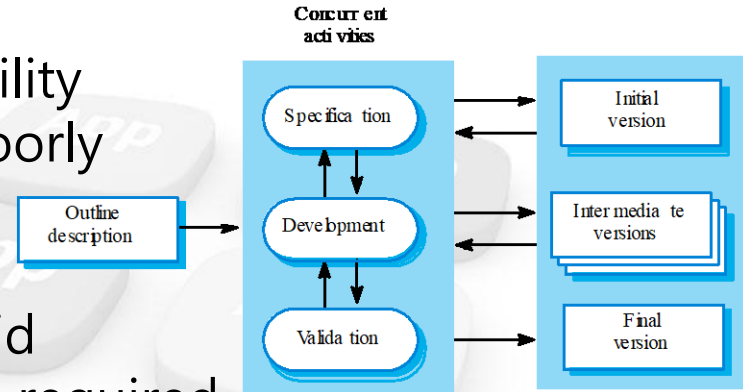
EVOLUTIONARY DEVELOPMENT

- Exploratory development
 - Objective is to work with customers and to evolve a final system from an initial outline specification. Should start with well-understood requirements and add new features as proposed by the customer
- Throw-away prototyping
 - Objective is to understand the system requirements. Should start with poorly understood requirements to clarify what is really needed

EVOLUTIONARY DEVELOPMENT

- Problems

- Lack of process visibility
- Systems are often poorly structured
- Special skills (e.g. in languages for rapid prototyping) may be required



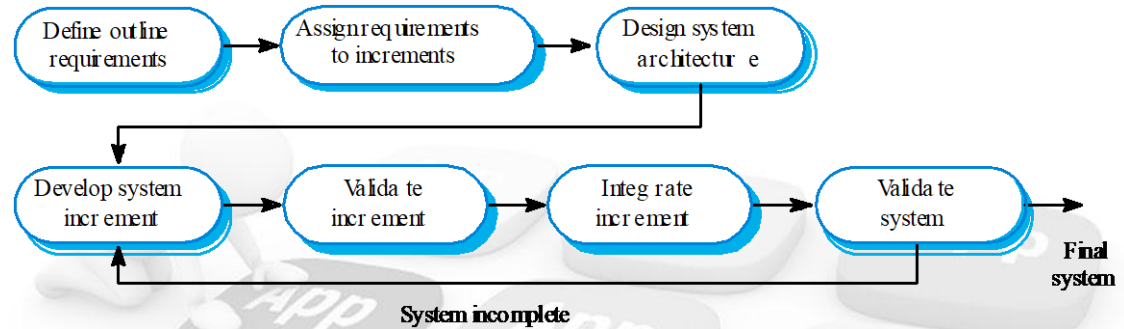
- Applicability

- For small or medium-size interactive systems
- For parts of large systems (e.g. the user interface)
- For short-lifetime systems

PROCESS ITERATION

- System requirements ALWAYS evolve in the course of a project so process iteration where earlier stages are reworked is always part of the process for large systems
- Iteration can be applied to any of the generic process models
- Two (related) approaches
 - Incremental delivery
 - Spiral development

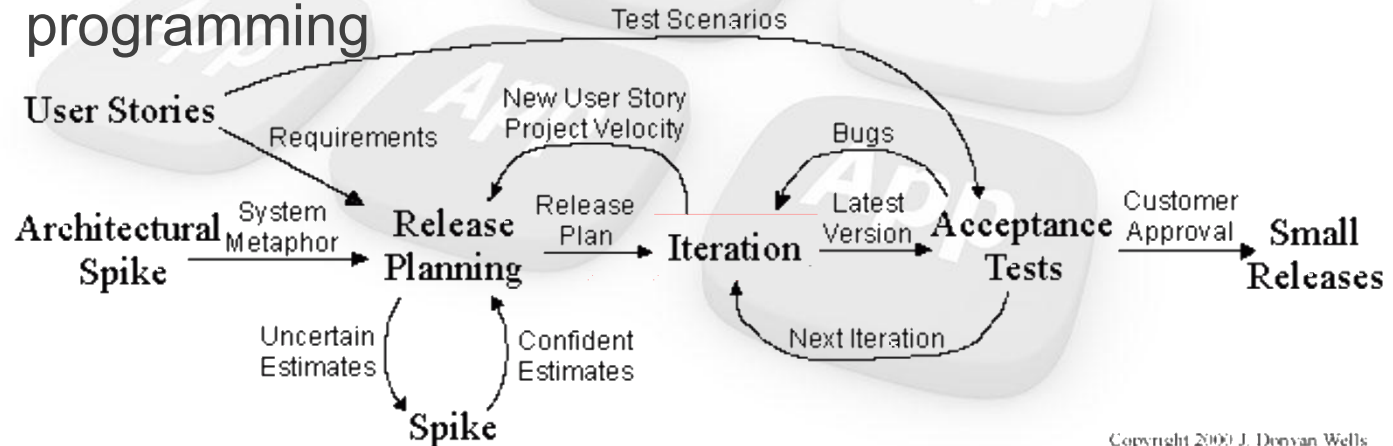
INCREMENTAL DELIVERY



- Rather than deliver the system as a single delivery, the development and delivery is broken down into increments with each increment delivering part of the required functionality
- User requirements are prioritised and the highest priority requirements are included in early increments
- Once the development of an increment is started, the requirements are frozen though requirements for later increments can continue to evolve

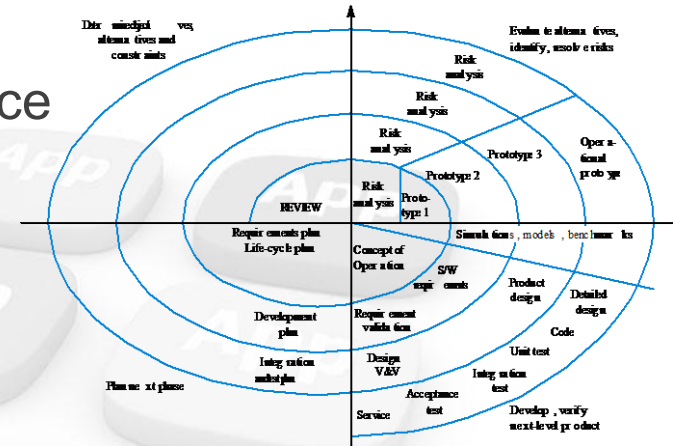
EXTREME PROGRAMMING

- An approach to development based on the development and delivery of very small increments of functionality
- Relies on constant code improvement, user involvement in the development team and pairwise programming



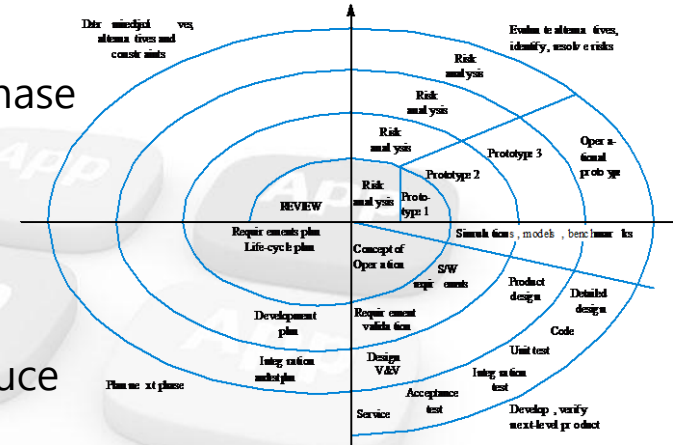
SPIRAL DEVELOPMENT

- Process is represented as a spiral rather than as a sequence of activities with backtracking
- Each loop in the spiral represents a phase in the process
- No fixed phases such as specification or design - loops in the spiral are chosen depending on what is required
- Risks are explicitly assessed and resolved throughout the process



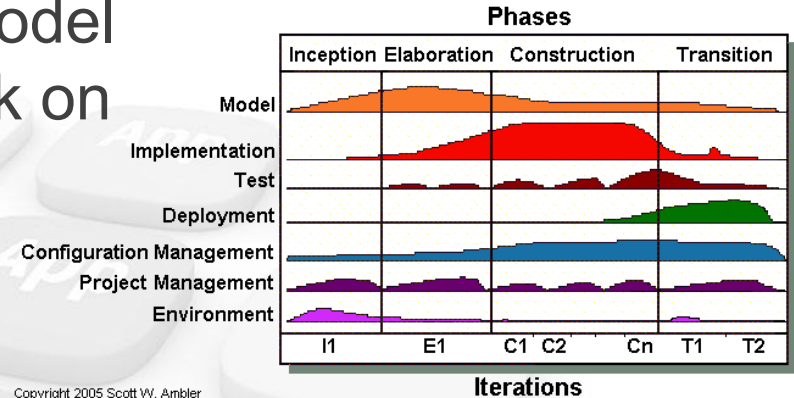
SPIRAL MODEL SECTORS

- Objective setting
 - Specific objectives for the phase are identified
- Risk assessment and reduction
 - Risks are assessed and activities put in place to reduce the key risks
- Development and validation
 - A development model for the system is chosen which can be any of the generic models
- Planning
 - The project is reviewed and the next phase of the spiral is planned



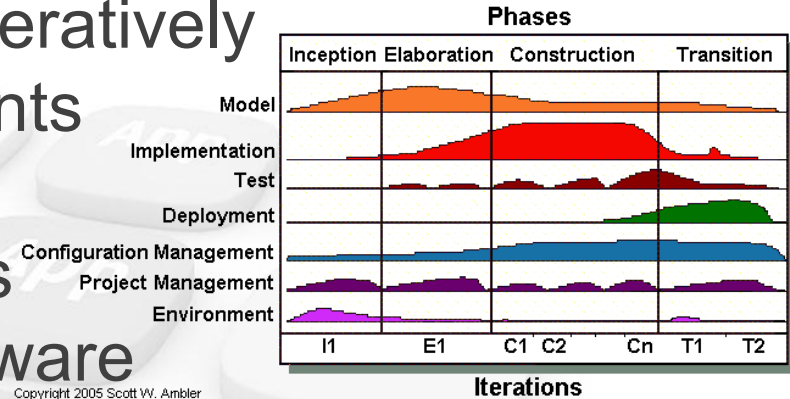
RATIONAL UNIFIED PROCESS (RUP)

- A modern process model derived from the work on the UML and associated process
- Normally described from 3 perspectives
 - A dynamic perspective that shows phases over time
 - A static perspective that shows process activities
 - A practice perspective that suggests good practice



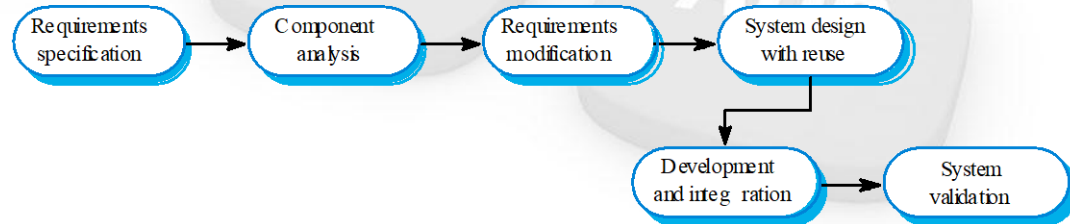
RUP GOOD PRACTICE

- Develop software iteratively
- Manage requirements
- Use component-based architectures
- Visually model software
- Verify software quality
- Control changes to software



INTEGRATION & CONFIGURATION DEVELOPMENT

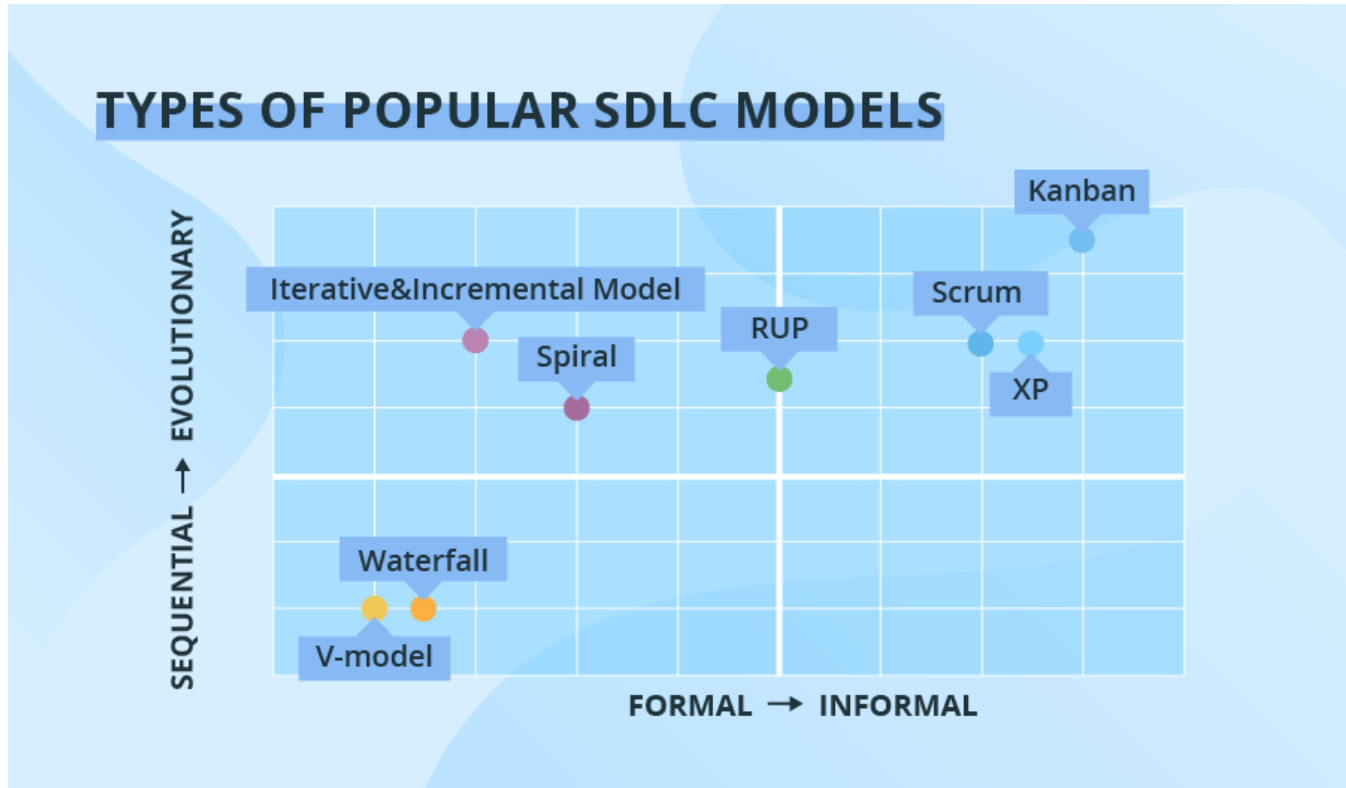
- Based on systematic reuse where systems are integrated from existing components or COTS (Commercial-off-the-shelf) systems
- Process stages
 - Component analysis
 - Requirements modification
 - System design with reuse
 - Development and integration
- This approach is becoming increasingly used as component standards have emerged



INTEGRATION & CONFIGURATION 2/2

- Types of software components:
 - **Web services** that are developed according to service standards and which are available for remote invocation.
 - Collections of objects that are developed as a **package/component** to be integrated with a component framework such as .NET or J2EE.
 - Stand-alone **commercial-off-the-shelf systems** (COTS) that are configured for use in a particular environment.

SUMMARY 1/5: SDLC OUTLINE



Source: Boris Shiklo, Chief Technology Officer, ScienceSoft (URL: <https://www.scnsoft.com/blog/software-development-models>)

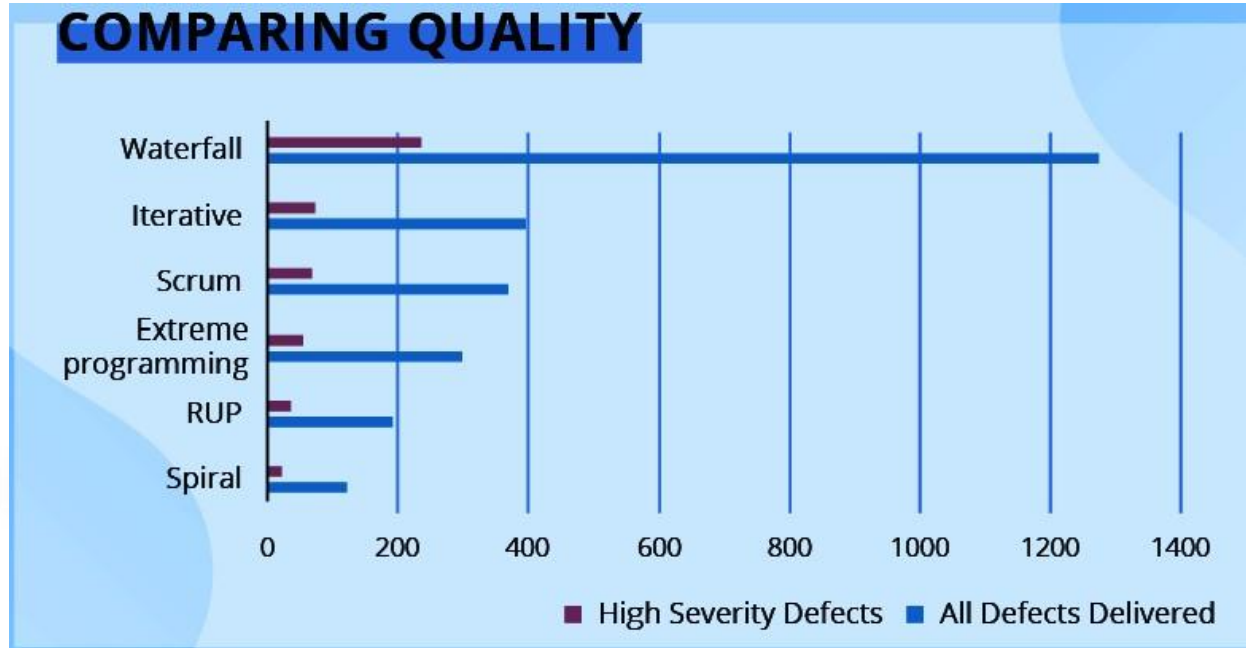
SUMMARY 2/5: COMPARING COST



Source: Boris Shiklo, Chief Technology Officer, ScienceSoft (URL: <https://www.scnsoft.com/blog/software-development-models>)

SUMMARY 3/5: COMPARING QUALITY

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Source: Boris Shiklo, Chief Technology Officer, ScienceSoft (URL: <https://www.scnsoft.com/blog/software-development-models>)

SUMMARY 4/5: COMPARING TCO 25/28

FOR 5 YEARS



Source: Boris Shiklo, Chief Technology Officer, ScienceSoft (URL: <https://www.scnsoft.com/blog/software-development-models>)

SUMMARY 5/5: SDLC SELECTION

Factors	Waterfall	Prototyping	Spiral	Incremental	Agile
Unclear User Requirement	Poor	Good	Excellent	Good	Excellent
Unfamiliar Technology	Poor	Excellent	Excellent	Good	Poor
Complex System	Good	Excellent	Excellent	Good	Poor
Reliable system	Good	Poor	Excellent	Good	Good
Short Time Schedule	Poor	Good	Poor	Excellent	Excellent
Strong Project Management	Excellent	Excellent	Excellent	Excellent	Excellent
Cost limitation	Poor	Poor	Poor	Excellent	Excellent
Visibility of Stakeholders	Good	Excellent	Excellent	Good	Excellent
Skills limitation	Good	Poor	Poor	Good	Poor
Documentation	Excellent	Good	Good	Excellent	Poor
Component reusability	Excellent	Poor	Poor	Excellent	Poor

Source: Mohamed Sami (URL: <https://melsatar.blog/2012/03/21/choosing-the-right-software-development-life-cycle-model/>)

KEY POINTS 1/2

- Software processes are the activities involved in producing and evolving a software system
- Software process models are abstract representations of these processes
- General activities are specification, design and implementation, validation and evolution
- Generic process models describe the organisation of software processes. Examples include the waterfall model, evolutionary development and component-based software engineering
- Iterative process models describe the software process as a cycle of activities

KEY POINTS 2/2

- Requirements engineering is the process of developing a software specification
- Design and implementation processes transform the specification to an executable program
- Validation involves checking that the system meets to its specification and user needs
- Evolution is concerned with modifying the system after it is in use
- CASE technology supports software process activities

THANK YOU...



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